

Course Abstract

The Cybersecurity Bootcamp for Professionals is a comprehensive, hands-on program designed to develop job-ready cyber talent equipped to operate in real-world, regulated environments. The curriculum blends foundational knowledge with advanced, practical application to ensure participants can contribute effectively from day one.

Participants build core capabilities across networking, Linux and Windows security, virtualization, containerization, and applied cryptography. These technical skills are complemented by training in security governance, risk management, and controls, enabling individuals to support audits, maintain compliance, and uphold enterprise security standards.

The program emphasizes real-world application through immersive, scenario-based learning. Participants develop skills in threat detection, incident response, fraud prevention, and AI-augmented security operations, leveraging data analytics tools such as Power BI and Copilot. Training scenarios are aligned to real banking environments, including fraud cases, business email compromise (BEC), wire transfer attacks, and emerging AI-driven threats.

A key differentiator of the program is its focus on the adversarial mindset—participants are trained to think like threat actors, enabling them to better anticipate, detect, and mitigate risks. This is reinforced through collaborative, role-based labs that simulate security operations center (SOC) workflows and promote teamwork, critical thinking, and communication.

The bootcamp employs an interactive, hands-on approach to ensure participants can apply theoretical knowledge in real-world scenarios. Key benefits include:

- **Practical Experience:** 70% of the course is dedicated to labs and simulations, allowing participants to develop and practice their skills in realistic settings.
- **Role-Based Learning:** One-third of labs require participants to work in security teams, adopting specific roles (e.g., Security Analyst, Network Engineer), enhancing collaboration and role-specific expertise.
- **Adversarial Mindset:** Participants learn to think like threat actors, fostering a proactive approach to identifying and mitigating security threats.
- **Comprehensive Coverage:** The course spans a wide range of topics, from basic security principles to advanced tools, preparing participants for diverse cybersecurity challenges.
- **Collaborative Learning:** Constant interaction and teamwork mirror real-world security operations, building essential communication and problem-solving skills.

Duration

10 Weeks (not inclusive of professional business skills)

Audience

This bootcamp is designed for professionals seeking to build or strengthen their cybersecurity skills. It is appropriate for individuals at the early-to-mid stage of their technology career who want to develop hands-on capabilities in security operations, threat detection, and incident response.

Prerequisites

Participants should have a basic familiarity with computers and operating systems. Prior experience with networking concepts or a command-line interface is helpful but not required.

Learning Objectives

By the end of this bootcamp, participants will be able to:

- Identify and mitigate security threats and vulnerabilities.
- Apply security measures to protect Linux and Windows systems.
- Implement and manage network security, including firewalls and authentication.

- Utilize cybersecurity tools for threat detection and incident response.
- Develop and enforce security policies and respond effectively to incidents.
- Collaborate in security teams to solve complex challenges.
- Explain core cryptographic concepts (symmetric/asymmetric encryption, hashing, PKI, digital certificates) and apply them to banking use cases such as TLS, data-at-rest encryption, tokenization, and key management.
- Use AI-assisted SOC tooling (e.g., Microsoft Security Copilot) to triage alerts, summarize incidents, and accelerate investigations while applying responsible-AI and model-risk guardrails appropriate to a regulated bank.
- Build security and fraud-monitoring dashboards in Power BI, including the use of Copilot for natural-language analysis, while applying data-governance guardrails (sensitivity labels, row-level security, DLP).

Course Topics

Week 1: Foundations of Networking and Security

- **Day 1-2: Fundamentals of Networking and Security Concepts**
 - o **Topics:** Introduction to Networking, Network Types, Data Transmission, CIA Triad, Security Basics
 - o **Mitigation Strategies:**
 - Implementing secure network protocols and encryption.
 - Segmenting networks to limit the spread of attacks.
 - o **Hands-On:** Identifying and Exploiting Network Vulnerabilities
 - o **Team-Based Lab:** Participants work in teams to secure a network infrastructure, with each team member adopting a specific role to ensure comprehensive security coverage.
- **Day 3-5: Introduction to Linux**
 - o **Topics:** Linux OS Overview, CLI Basics, Linux Directory Structure, File Permissions
 - o **Mitigation Strategies:**
 - Applying least privilege principles.
 - Regularly updating and patching Linux systems.
 - o **Hands-On:** Securing Linux Environments Against Common Attacks

Week 2: Advanced Linux and Windows Administration

- **Day 1: Introduction to Linux Security**
 - o **Topics:** Linux Security Principles, Linux Security Model, Common Vulnerabilities
 - o **Mitigation Strategies:**
 - Implementing security policies and regular compliance checks.
 - o **Hands-On:** Implementing Initial Security Policies
- **Day 2: System Hardening**
 - o **Topics:** System Updates, Disabling Unused Services, Kernel Hardening, SELinux, AppArmor
 - o **Mitigation Strategies:**
 - Regularly applying system updates and patches.
 - Disabling unnecessary services and ports.
 - o **Hands-On:** Hardening a Linux System
- **Day 3: User and Permission Management**
 - o **Topics:** User Account Security, File Permissions, ACLs
 - o **Mitigation Strategies:**
 - Implementing strong password policies and multi-factor authentication.
 - Regularly auditing user permissions and access controls.
 - o **Hands-On:** Securing User Accounts and File Permissions
 - o **Team-Based Lab:** Teams configure and enforce security policies for a Linux system, with each member focusing on specific areas such as user management, file permissions, and system

hardening.

- **Day 4-5: Windows Server Administration**
 - o **Topics:** Configuring Windows Servers, Managing Users and Policies, Windows BitLocker
 - o **Mitigation Strategies:**
 - Enforcing group policies to secure user accounts and system settings.
 - Using encryption tools like BitLocker to protect sensitive data.
 - o **Hands-On:** Hardening Windows Servers

Week 3: Virtualization and Container Foundations

- **Day 1-2: Introduction to Virtualization**
 - o **Topics:** Virtual Machines, Supervisor vs. Hypervisor, Client-Side and Server-Side Virtualization
 - o **Mitigation Strategies:**
 - Isolating virtual machines to prevent lateral movement in case of a breach.
 - Regularly updating and securing hypervisor software.
 - o **Hands-On:** Identifying and Mitigating Virtualization-Specific Threats
- **Day 3-5: Enterprise Virtualization**
 - o **Topics:** Type 1 Hypervisors, Enterprise-Level Virtualization Management
 - o **Mitigation Strategies:**
 - Implementing secure access controls for managing virtualized environments.
 - Monitoring virtual machine activity for signs of compromise.
 - o **Hands-On:** Securing Enterprise Virtualization Environments
 - o **Team-Based Lab:** Teams are tasked with securing a virtualized enterprise environment, with each member adopting a role focused on specific aspects of virtualization security, such as access control, monitoring, and incident response.

Week 4: Network Security and Cryptography 101

- **Day 1-2: Networking Protocols, Authentication and Authorization**
 - o **Topics:** Routing and Internet Protocols, DHCP, Securing Network Components, Authentication Vulnerabilities, OAuth2, JSON Web Tokens (JWT), Multi-Factor Authentication (MFA), Identity & Access Management (IAM) fundamentals
 - o **Mitigation Strategies:**
 - Implementing secure routing protocols (e.g., OSPF with authentication).
 - Using DHCP snooping to protect against IP spoofing.
 - Enforcing strong authentication policies and MFA across bank systems.
 - o **Hands-On:** Configuring Networks to Defend Against Protocol-Based Attacks; Strengthening Authentication Mechanisms.
- **Day 3: Cryptography Fundamentals**
 - o **Topics:** Why cryptography matters in a bank (confidentiality, integrity, authentication, non-repudiation); symmetric vs. asymmetric encryption; AES, RSA, and ECC; cryptographic hashing (SHA-2, SHA-3) and message authentication codes (HMAC); digital signatures.
 - o **Hands-On:** Encrypting and decrypting files with symmetric (AES) and asymmetric (RSA) algorithms; generating and verifying SHA-256 hashes and digital signatures.
- **Day 4: Public Key Infrastructure (PKI) and Certificates**
 - o **Topics:** Public and private key pairs; X.509 digital certificates; Certificate Authorities (CAs) and the chain of trust; certificate lifecycle (issuance, renewal, revocation, OCSP/CRL); TLS/SSL handshake; HTTPS in depth; common certificate-related attacks (rogue CAs, expired/misissued certs).
 - o **Hands-On:** Generating a CSR, issuing certificates from an internal CA, configuring TLS on a sample web service, inspecting certificates in a browser and with OpenSSL.
- **Day 5: Applied Cryptography for Banking**

- o **Topics:** Encryption at rest vs. in transit; key management and Hardware Security Modules (HSMs); tokenization vs. encryption for cardholder data (PCI DSS 4.0 considerations); database and disk encryption (TDE, BitLocker); secure key storage in cloud (AWS KMS, Azure Key Vault); awareness of NIST Post-Quantum Cryptography standards (FIPS 203 ML-KEM, FIPS 204 ML-DSA, FIPS 205 SLH-DSA) and "harvest now, decrypt later" risk for long-lived bank data.
- o **Mitigation Strategies:**
 - Selecting appropriate algorithms and key sizes for bank workloads.
 - Protecting keys through HSMs, vaulting, and rotation.
 - Tracking PQC migration roadmaps for high-value data.
- o **Team-Based Lab:** A junior SOC team is asked to investigate a TLS misconfiguration alert on a customer-facing banking app, identify weak ciphers and certificate issues, and recommend remediation.

Week 5: Next Generation Firewalls

- **Day 1-3:** Firewall Technologies
 - o **Topics:** Traditional vs. Next Generation Firewalls, Firewall Rules, Virtual Appliances
 - o **Mitigation Strategies:**
 - Regularly updating firewall rules based on emerging threats.
 - Segmenting networks and applying different security policies per segment.
 - o **Hands-On:** Configuring Firewalls to Block Advanced Threats
 - o **Team-Based Lab:** Teams configure and manage next-generation firewalls, with each participant responsible for different aspects such as rule configuration, traffic monitoring, and network segmentation.
- **Day 4-5:** Enterprise Security Posture
 - o **Topics:** Multi-LAN Management, Web Access, FTP Security, Antivirus Configuration
 - o **Mitigation Strategies:**
 - Implementing web content filtering to block malicious sites.
 - Using antivirus software with real-time scanning to detect and mitigate threats.
 - o **Hands-On:** Developing a Robust Security Posture

Week 6: Cybersecurity Threats and Vulnerabilities

- **Day 1-2:** Threat and Vulnerability Management
 - o **Topics:** Security Controls, Threat Intelligence, Incident Monitoring
 - o **Mitigation Strategies:**
 - Conducting regular vulnerability scans and assessments.
 - Utilizing threat intelligence feeds to stay ahead of emerging threats.
 - o **Hands-On:** Conducting Vulnerability Assessments
- **Day 3-5:** Software and Systems Security
 - o **Topics:** Security Monitoring, Digital Forensics, Indicator Analysis
 - o **Mitigation Strategies:**
 - Implementing endpoint protection to monitor and respond to threats.
 - Applying forensic techniques to identify root causes of security incidents.
 - o **Hands-On:** Applying Forensic Techniques

Week 7: Security Operations, Incident Response and Integration of AI

- **Day 1-2:** Security Operations and SOC Workflows
 - o **Topics:** Tiered SOC model (T1 Monitoring & Triage, T2 Investigation, T3 Hunt/IR), alert triage process, escalation paths, Incident Response Procedures, Risk Mitigation, SIEM Alerts, runbooks

- and playbooks for common bank alerts (failed logins, impossible travel, BEC indicators, suspicious wire-transfer activity).
- o **Mitigation Strategies:**
 - Developing and testing incident response plans regularly.
 - Monitoring SIEM alerts for early detection of threats.
 - o **Hands-On:** Triageing a queue of simulated bank SIEM alerts as a Tier 1 analyst, properly escalating to Tier 2; simulating incident response.
 - o **Team-Based Lab:** Teams work together to respond to a simulated security incident, with roles such as incident commander, forensic analyst, and remediation specialist.
- **Day 3: Integration of AI in the SOC — Foundations**
 - o **Topics:** Basic AI awareness for analysts (what LLMs are and are not); how AI is used across the SOC (alert triage, summarization, query generation, threat-intel synthesis); overview of Microsoft Security Copilot and the Defender XDR Alert Triage and Phishing Triage agents; building confidence in interpreting AI outputs, including verifying citations, evidence, and reasoning before acting.
 - o **Hands-On:** Walking through an AI-generated alert verdict, validating each piece of supporting evidence, and deciding whether to accept, modify, or escalate.
 - **Day 4: AI-Assisted Triage and Prompting Skills**
 - o **Topics:** Entry-level prompting skills for SOC analysts (clear context, specific intent, follow-up questions); using AI to summarize incidents, draft KQL/SPL queries, explain malware behavior, and produce incident write-ups; using AI in cloud monitoring contexts (Defender for Cloud, Sentinel); awareness of AI + automation (SOAR) workflows.
 - o **Hands-On:** Using a Security Copilot-style assistant (or equivalent) to triage a phishing alert, summarize a multi-step incident, and draft a stakeholder communication.
 - **Day 5: Responsible AI in a Regulated Bank**
 - o **Topics:** AI-driven threats analysts will face (AI-crafted phishing, deepfake-enabled social engineering, prompt injection); guardrails for AI use in a bank — data handling, prompt hygiene, avoiding sensitive data in prompts; awareness of the regulatory landscape, including the Treasury Financial Services AI Risk Management Framework and OCC/Fed/FDIC revised Model Risk Management guidance; when to trust, verify, or override AI output; compliance and assessment basics (data privacy, GDPR/GLBA/NYDFS Part 500 awareness, cloud audits).
 - o **Hands-On:** Reviewing an AI-generated investigation, identifying where the AI was wrong or unclear, and documenting an analyst-level override.

Week 8: SOC Tools and Data Analytics 101

- **Day 1: Packet and Protocol Analysis**
 - o **Topics:** Network Sniffing Tools, Data Manipulation, PCAP File Analysis
 - o **Mitigation Strategies:**
 - Analyzing network traffic to detect anomalies and potential threats.
 - Implementing intrusion detection systems (IDS) to monitor network traffic.
 - o **Hands-On:** Traffic Flow Analysis
- **Day 2: SIEM Deep Dive and the Modern Bank SOC Stack**
 - o **Topics:** SIEM concepts and architecture; hands-on with Splunk for log analysis and detection; overview of Microsoft Sentinel and the bank-relevant Defender XDR / Security Copilot integration; how SIEM, EDR, SOAR, and AI assistants fit together.
 - o **Hands-On:** Writing basic Splunk searches and Sentinel KQL queries to investigate a simulated banking incident.
- **Day 3: Data Analytics 101 for Security and Fraud**
 - o **Topics:** Why data analytics matters in a bank SOC (operational metrics, fraud trends, regulatory reporting); structured vs. unstructured data; common security/fraud datasets (auth logs, transaction logs, alert outcomes); data visualization principles; what makes a good SOC or fraud dashboard (MTTD, MTTR, alert volume, false-positive rate, top tactics, fraud loss).

- o **Hands-On:** Exploring a sample bank security/fraud dataset and identifying questions a dashboard should answer.
- **Day 4:** Power BI Fundamentals for Security Analysts
 - o **Topics:** Power BI Desktop and Service overview; connecting to data sources (CSV, SQL, log exports); shaping data with Power Query; building visuals (bar, line, KPI cards, maps); slicers and filters; publishing and sharing reports; introduction to data modeling and simple DAX measures.
 - o **Hands-On:** Building a SOC Operations dashboard from sample data — alerts by severity, MTTR by tier, top alert types, false-positive ratio.
- **Day 5:** Power BI Copilot and Dashboards for Fraud and Governance
 - o **Topics:** Using Copilot in Power BI to summarize reports, generate visuals from natural-language prompts, and draft DAX/measures; governance and guardrails for Copilot in regulated industries (sensitivity labels, DLP policies, row-level security, audit logging, approved use cases); building a banking-relevant fraud dashboard (suspicious transactions, BEC indicators, geographic anomalies, watchlist hits).
 - o **Mitigation Strategies:**
 - Integrating SOC tools into a single triage workflow.
 - Using Copilot to accelerate, not replace, analyst judgment.
 - Applying DLP and sensitivity labels to all Power BI assets handling regulated bank data.
 - o **Team-Based Lab:** Teams build and present a bank-flavored security/fraud dashboard, using Copilot to generate at least one visualization and one written summary, then defend their dashboard design choices and governance controls to a stakeholder panel.

Week 9: Social Engineering and Human Factor Security

- **Day 1-2:** Social Engineering Tactics
 - o **Topics:** Understanding Social Engineering Threats, Phishing, Pretexting, Baiting, and Tailgating
 - o **Mitigation Strategies:**
 - Training and Awareness Programs
 - Implementing Multi-Factor Authentication (MFA)
 - Creating a Culture of Security Vigilance
 - Using Technical Controls (Email Filtering, Anti-Phishing Tools)
 - o **Hands-On:**
 - Crafting Social Engineering Scenarios
 - Developing and Implementing Mitigation Plans
 - Testing Mitigation Strategies in a Simulated Environment
- **Day 3-5:** Enterprise-Level Security Considerations
 - o **Topics:** Organizational Security Policies, Securing Communication Channels
 - o **Mitigation Strategies:**
 - Policy Enforcement and Regular Security Audits
 - Secure Communication Protocols and Encryption
 - Incident Response Planning for Social Engineering Attacks
 - o **Hands-On:**
 - Developing Security Awareness Programs
 - Conducting Security Audits Focused on Human Factors
 - Simulating Incident Response for a Social Engineering Attack

Week 10: Capstone Project and Review

- **Day 1-3:** Capstone Project
 - o **Project:** Comprehensive Project Integrating All Learned Skills, Securing a Complex Environment
 - o **Mitigation Strategies:**
 - Applying all mitigation techniques learned throughout the course.
 - Demonstrating a full incident response, from detection to remediation.
- **Day 4-5:** Review and Presentation

- o **Activities:** Presentations, Peer Review, Final Review of Key Concepts

Professional Business Skills

Topics can be selected based on need. Business skills can either be presented as part of bootcamp or during on the-job period as part of continuous learning.

Personal Development

Personality Assessment
Psychological Safety
Growth Mindset
Emotional Intelligence
Crash Course: Productivity and Time Management

Culture and the Team

Team Vision, Mission and Values
Managing team conflict
Celebrating Failures
Meeting Facilitation

Communication

Written and Verbal Communication
Asking Better Questions
Managing Difficult Conversations
Giving and Receiving Feedback

Introduction to Design Thinking

Certification Pathways

This bootcamp is not a certification class; however, it equips participants with the knowledge and experience to help them pursue certification in the following areas:

1. Certified Information Systems Security Professional (CISSP)
2. Certified Ethical Hacker (CEH)
3. CompTIA Security+
4. Certified Information Security Manager (CISM)
5. Certified Information Systems Auditor (CISA)
6. GIAC Security Essentials (GSEC)
7. Certified Incident Handler (ECIH)
8. Certified Cloud Security Professional (CCSP)
9. Offensive Security Certified Professional (OSCP)
10. Certified in Risk and Information Systems Control (CRISC)

Course Code

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